Sim2Win: A Team-Agnostic, Event-Based Pre-Match Tactical Recommendation System for Football

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ABSTRACT & KEYWORDS

Sim2Win addresses the gap between descriptive outcome prediction and prescriptive tactical recommendation. By utilizing team-agnostic rolling tactical profiles, the system predicts match out-comes for unseen teams, outperforming established rating systems like ELO-G and Pi-Rating.

Keywords: Machine Learning, Sports Analytics, Prescriptive Analytics, Tactical Profiling.

BACKGROUND

The central operational challenge in professional football is pre-match tactical preparation. Coaching staff must decide on formations and pressing intensity before each match. Existing data-driven approaches rely on team identity features, which fail for newly promoted clubs or national teams.

PROBLEM STATEMENT

Can a machine learning system generate reliable pre-match tactical recommendations for teams it has never encountered during training by using only tactical behavioral features derived from event-based data?

RESEARCH QUESTION

To what extent do rolling tactical feature profiles enable generalizable match outcome prediction and actionable tactical recommendation for teams absent from the training distribution?

INTERDISCIPLINARY FRAMEWORK

Leading Discipline: Computer Science

Focuses on supervised classification, unsupervised clustering (K-Means), and rigorous out-of-distribution evaluation protocols (LOCO).

Helping Discipline: Sports Science

Provides domain logic for feature engineering (e.g., pressing efficiency, xG per shot) and tactical performance indicators.

HYPOTHESIS

Tactical profile vectors carry sufficient predictive signal to enable a CatBoost classifier to outperform identity-based systems on unseen teams because they capture universal stylistic behavioral regularities.

METHODOLOGY

- **Data Collection:** 1,411 team-match rows from 11 professional competitions (StatsBomb Open Data).
- **Feature Engineering:** 5-match rolling windows featuring ratios like Pressing Efficiency and Chaos Index.
- **Clustering:** K-Means ($K = 8$) to identify discrete playstyle archetypes (e.g., high-press, counterattack).
- **Evaluation:** Leave-One-Competition-Out (LOCO) protocol.

KEY FINDINGS & RESULTS

Metric	Sim2Win Performance
Mean ROC-AUC	**0.704**
Tactical Agreement	**60.9%**
Win Pred. Accuracy	**67.4%**

Sim2Win outperformed ELO-G and Pi-Rating baselines on every competition and every metric. Pressing intensity and shot quality emerged as the dominant predictors.

RECOMMENDATIONS & CONCLUSION

Recommendations: Teams should prioritize high-fidelity behavioral data over historical reputation when preparing for unfamiliar opponents.

Conclusion: Sim2Win demonstrates that behavioral" DNA" is a more robust predictor of success than historical identity in dynamic competitive environments.

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